# BLOCK 2

## MODERN THEORIES OF INTERNATIONAL TRADE

# BLOCK INTRODUCTION

This block presents the modern theories of International Trade. It discusses the need, rational and measurement of Intra-Industry Trade (IIT). This block also elucidates the extensions of classical trade theories with more realistic assumptions which explain the current patterns of international trade in the world market. This block comprises of two units.

**Unit 3 Intra-Industry Trade (IIT)** deliberates on the phenomena of intra-industry trade and identifies the need for a different theory for intra-industry trade. The two types of IIT – horizontal and vertical IIT are also elaborated in this Unit. The unit also explained 'love of variety' and 'ideal variety' approach in the Intra-Industry trade. Possible reasons behind IIT in identical products are also discussed in this unit. The unit also presents the Grubel Lloyd (GL) index to measure the intensity of IIT.

**Unit 4 Alternative Explanations of International Trade** Theories sheds light on new, complementary trade theories based on real world situations using the concepts of Economies of Scale, Imperfect Competition, Product differentiation and Technological Gap. This unit probe the effect of relaxing each of the classical and Heckscher-Ohlin trade theory assumptions on International trade. It critically examines the impact of Logistics cost and environmental standards on the world trade.

# UNIT 3 INTRA-INDUSTRY TRADE *

## Structure



## 3.0 OBJECTIVES

After studying this Unit, you should be able to:

- describe the meaning of Intra-Industry trade;
- distinguish between horizontal and vertical Intra-Industry trade;
- identify the need for a separate theory for Intra-Industry trade;
- explain the 'love of variety' and 'ideal variety' approach in Intra-Industry trade;
- elucidate Intra-Industry trade in Horizontally and vertically differentiated Commodities;
- list the major problems in measurement of intensity of Intra-Industry trade; and
- measure intensity of Intra-Industry trade using the Grubel Lloyd (GL) index.

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## 3.1 INTRODUCTION

We have already discussed in the previous unit about the gains from international trade i.e., the various benefits a country derives from trading goods and services with other countries. Each trading country gains when the total output increases due to the division of labour and specialisation. This unit will discuss the phenomena of intra-industry trade and why there is a need for a different theory for intra-industry trade. Greater specialisation in the manufacturing of unique varieties or product lines by individual plants in different countries has led to the tremendous growth of IIT. In order to exploit economies of scale in production, individual plants specialize in unique variety and thus leading to growth of IIT.

Further, greater specialization at the level of distinct components used in an industry has promoted IIT in intermediate goods. In this unit, we will also discuss about the two types of IIT – horizontal and vertical IIT. Horizontal IIT is the simultaneous export and import of differentiated products within an industry of a similar quality. Vertical IIT, on the other hand, refers to the exchange of differentiated products of different qualities.

We will also discuss 'love of variety' and 'ideal variety' approach in the Intra-Industry trade. We will further discuss the possible reasons behind IIT in identical products.

Two major issues in the empirical analysis of IIT are concerned with: i) the definition of 'industry' and ii) measurement of IIT. We will discuss about these issues along with the Grubel Lloyd (GL) index to measure the intensity of IIT.

## 3.2 TRADE LIBERALIZATION AND THE PHENOMENON OF INTRA-INDUSTRY TRADE

Mercantilism Trade liberalisation is expected to rationalise the choice of product lines by individual plants in domestic industries. Production and export of all varieties in a given industry become impossible if economies of scale in production are to be reaped. Suppose the products in an industry are differentiated and each is manufactured with increasing returns to scale. In that case, a country may specialise in producing a subset of varieties for home consumption and export, while importing those varieties that are not supplied domestically. In this context, it becomes important to understand the structure of exports in its relation to imports within the industry. For understanding this dimension of trade pattern, trade economists have evolved the concept of intra-industry trade (henceforth IIT), which means the **simultaneous occurrence of exports and imports within the same industry**. To be more precise, IIT means the simultaneous export and import of products, which are very close substitutes for each other in terms of factor inputs and consumption.

Growth of IIT is an outcome of greater specialisation in the manufacturing of unique varieties or product lines by the individual plants in different countries.

Because there are fixed costs associated with the production of each variety, it is imperative that each plant specializes in a unique variety if economies of scale in production are to be exploited. Therefore, the firms in each country would manufacture a subset of the varieties within an industry for meeting home demand and export. At the same time, the varieties that are not produced within the country will be imported to meet the domestic demand. This leads to the occurrence of IIT in final consumer goods. Further, greater specialization at the level of distinct components used in an industry will promote IIT in intermediate goods.

The phenomenon of IIT began to gain sustained interest since the early 1960s, after the formation of the European Economic Community (EEC). The conventional wisdom is that competition would induce a process of resource reallocation from the import competing industries to those industries where the country has comparative advantages. It follows that a natural consequence of trade liberalization is the expansion of inter industry trade – that is, export increase from one set of industries and import increase in another. Further, it is held that trade liberalization invariably involves adjustment costs as some domestic industries may go out of business.

However, a number of studies in the context of different countries suggest that trade liberalization biases the country's trade expansion towards IIT rather than inter industry trade. In other words, the pattern of specialisation was not the one whereby different countries specialised in different products. Instead, different countries tended to specialise in different types of a given product and there was an expansion of exports from practically every industry rather than the demise of inefficient manufacturing industries.

In a liberalized environment, different countries would specialise in different types of a given product and there could be expansion of exports from practically every industry. The apprehension that import liberalisation might lead to the demise of domestic industries in the developing countries (the fear of de-industrialisation) becomes untenable if we indeed find that the intensity of IIT increases with the reduction of trade barriers. The widely held 'smooth adjustment hypothesis' states that a high share IIT will be associated with relatively low adjustment costs, since, with intra-industry adjustment, workers move within industries rather than between them.

There is the distinction between the two types of IIT – horizontal and vertical IIT. Horizontal IIT refers to the simultaneous export and import of differentiated products within an industry of a similar quality. Vertical IIT, on the other hand, refers to the exchange of differentiated products that are of different qualities. Horizontal IIT arises when countries with similar factor endowment ratios specialize in unique varieties to exploit internal scale economies. Vertical IIT, on the other hand, arises when countries with different factor endowment ratios specialize in varieties that are differentiated by quality: capital-abundant countries would specialize in the production of higher quality varieties while labor-abundant countries would specialize in low quality varieties. Vertical IIT between two countries will arise given an overlap in the demand for different qualities. This possibility is enhanced by the greater factor endowment difference between the countries. In practice, the trade that emerges from specialization at different stages of producing a particular good may also be treated as vertical IIT (for example, China imports parts of a product and exports the finished product). The extent of adjustment costs associated with trade liberalization can differ depending on whether IIT is horizontal or vertical. It is generally held that while any type of IIT entails relatively lower adjustment costs than inter-industry trade, the adjustment cost will be the least when IIT is horizontal in nature.

## Check Your Progress 1

**Note:** i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Describe Intra-Industry trade.

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2) Differentiate between horizontal and vertical IIT?

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3) Discuss the impact of trade liberalization on growth of IIT in a country.

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## 3.3 THEORY OF INTRA-INDUSTRY TRADE

### 3.3.1 Need for a Separate Theory for Intra-Industry trade

The early empirical studies on IIT stimulated analyses that were directed at establishing the extent of the phenomenon in different countries at different points in time. Grubel and Lloyd (1975) are noteworthy among these works, which documents detailed evidence of IIT for all major industrialised countries. While Grubel and Lloyd (1975) suggested a number of plausible explanations for IIT, it remained `a phenomenon in search of a theory' till the late 1970s.

'What determines the pattern of international trade?' This question has been a subject of considerable interest to economists ever since the formulation of comparative advantage theory by David Ricardo. The well-known Heckscher-Ohlin-Samuelson (H-O-S) model, the synthesis of the initial attempts, remained the main theory for quite a long time. This model provides explanation for the emergence of inter-industry trade based on factor endowment difference of countries. This theory's core is the proposition that trade flows between countries are in `complementary' goods with differing factor intensities. Accordingly, the bulk of empirical testing since then has been on exploring such complementarity.

Ever since Leontief's results cast doubt upon the H-O-S explanations (the well-known Leontief paradox), there has been considerable work on refining the tests and restating the models to make them more suitable for testing. The scepticism on the explanatory power of the H-O-S model originated from the Leontief paradox apart, the new attempts have been stimulated by yet another observation. That is, a predominant amount of international trade takes place among industrial nations, with relatively similar resource endowments, which is difficult to reconcile with the prediction of the H-O-S model. It was generally held that the traditional trade models, based on the assumption of perfect competition cannot provide a proper understanding of IIT, particularly among the industrial countries.

Clearly, what appeared to be imperative was a general equilibrium theory of international trade, which incorporates the elements of economies of scale, product differentiation and imperfect competition. The pursuit of the task, however, was less easy than the stating of it. A major difficulty was that scale economies internal to firms are incompatible with perfect competition that was assumed in all traditional models. In short, theorists in the field lacked analytic tools necessary to formalise and refine the ideas in tractable mathematical models.

However, some major developments in economic theory in the 1970s completely changed the scenario. First, there was substantial progress in the theoretical modelling of imperfect competition, which led to the revival of Chamberlin's "large group" analysis of competition between similar firms producing differentiated products. Secondly, the later half of 1970s witnessed rigorous theoretical analysis of the process of product differentiation, by Spence (1976) and Dixit and Stiglitz (1977). This approach, called the "love-of variety approach" considers that each consumer has a taste for many different varieties. Later, Lancaster (1979) advanced an alternative approach, known as the "ideal variety approach", which considers that each consumer has a desire for the most preferred variety rather than many varieties. Though there is a significant difference in the two approaches, both represent cases of what is called "horizontal differentiation".

These developments provided economists with appropriate toolkit to facilitate the incorporation of product differentiation, economies of scale and imperfect competition into the formal trade theory and proliferation of theoretical research on IIT. The new approach proved useful for providing theoretically convincing explanations for IIT and integrating them with the conventional explanations for inter-industry trade based on factor proportions. It may, however, be noted that as long as a general theory of imperfect competition does not exist, it is not possible to arrive at a general theory of trade with imperfect competition. There are too many ways in which imperfect market structure can be modelled and in principle there can be as many models of specialisation and trade.

The theoretical literature may be grouped as belonging to four different strands. The earlier attempts represent models of horizontal IIT – that is, the exchange of commodities differentiated by those attributes excluding quality. Horizontal IIT is explained by economies of scale in the presence of product differentiation and imperfect competition. Such models have greater relevance for understanding the occurrence of IIT among developed countries. Significant attempts were made in the later years to offer a different strand of analysis, which represent cases of vertical IIT – that is, the exchange of commodities differentiated by quality. In general, these models explain the pattern of IIT along the lines similar to the pattern of inter-industry trade predicted in the conventional trade model, according the central role to factor endowment differences. Such models are considered to be particularly relevant to explain the presence of IIT between unequal trade partners. A third strand of analysis attempted to explain IIT in differentiated intermediate products in the presence of international returns to scale. Finally, some theoretical attempts have aimed at providing explanations for IIT in identical commodities on the basis of strategic interactions of internationally oligopolistic firms. In what follows, we shall sum up the essential elements of the major strands of theories.

## Check Your Progress 2

**Note:** i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) What led to need for a Separate Theory for Intra-Industry trade?

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2) Describe ‘love of variety approach’ in intra industry trade.

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3) Describe 'ideal variety approach' in intra industry trade.

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## 3.3.2. IIT in Horizontally Differentiated Commodities.

Theoretical models of IIT in horizontally differentiated commodities are generally based on the Chamberlinian notion of monopolistic competition. As regards the treatment of product differentiation, the love-of-variety approach and the ideal variety approach are used by various economists. Notwithstanding differences in the approach followed to specify the demand side, all these models yielded broadly similar results. The essential arguments of these models may be summed up as follows.

In Krugman's model, the love of variety approach is assumed on the demand side. Each consumer has the same utility function, wherein all varieties enter symmetrically. The utility function for each of the (identical) consumer is:

$$U = \sum_{i=1}^{N} v(c_i), \qquad v' > 0, \qquad v'' < 0 \qquad (1)$$

Where $c_i$ is the consumption of each variety i = 1,….N. Given the assumption of diminishing marginal utility from each variety $\left(v'' < 0\right)$, aggregate utility (U) can be maximised by increasing the number of varieties (N) consumed rather than by spending the income on few varieties.

The economy under consideration is assumed to have only one factor of production - that is, labour. The underlying market structure is assumed to be the Chamberlinian monopolisitc competition. A large number of firms exist, each of which produces a different variety ($y_i$) of the same good Y, but with the same cost function:

$$L_i = \alpha + \beta y_i \qquad (2)$$

$\alpha$ stands for fixed labor input while $\beta$ is marginal labor input. The term $\alpha$ incorporates economies of scale into the model. In this model, each firm produces a unique variety as no two firms have incentive to produce the same variety. This is due to the fact that economies of scale operate at the level of varieties and, at the same time, all available varieties enter the consumer's utility function.

The marginal cost, $MC_i$ for the firm is $w\beta$ while the average cost is given as:

$$AC_i = \frac{wL_i}{y_i} = \frac{w(\alpha + \beta y_i)}{y_i} = \frac{w\alpha}{y_i} + w\beta \qquad (3)$$

This means that AC falls with the increase in output ($y_i$) while the MC is constant (see Figure 3.1).

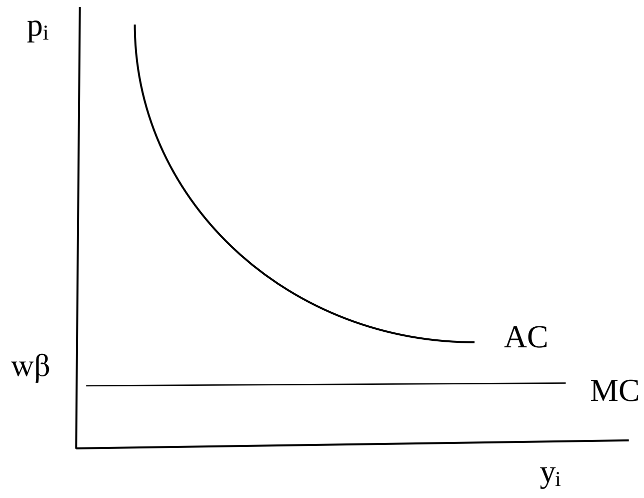


**Figure 3.1: Average Cost Curve and Marginal Cost curve**

The symmetry assumption in regard to both cost and demand functions will ensure that all varieties actually produced will be produced in the same quantity and at the same price. Thus, to arrive at equilibrium, one needs to determine only the price and output of a single representative variety (therefore, we can ignore the subscript *i*). The two equilibrium conditions for firms under monopolistic competition are: (i) MR=MC and (ii) p = AC (zero profit in the long run), where MR is marginal revenue and *p* is price. When η is the elasticity of demand for variety i. MR equals $\frac{p}{1}\left(1-\frac{1}{\eta}\right)$

Using MR=MC condition, we can show that:

$$\frac{p}{w} = \beta\left(\frac{\eta}{\eta-1}\right) \qquad (4)$$

Similarly, using the p = AC condition, it can be shown that

$$\frac{p}{w} = \left(\frac{\alpha}{Lc}\right) + \beta \qquad (5)$$

Where η is the elasticity of demand for variety i. We assume that $d\eta_i/dc_i < 0$ - that is, elasticity of demand decreases with increase in consumption of a variety (note that this assumption always hold for any linear demand curve, q = a - bp). With this assumption, it is clear that the graph of equation 4 (AB curve) slopes upward with the increase in c (see Figure 3.2). This is because increase in c leads to a decline in η, and therefore the term $\eta/\eta-1$ in equation (4) increases as β is a constant. Turning to equation 5, the graph of this equation (PQ curve) slopes downward with the increase in c (this is firm's average cost curve).

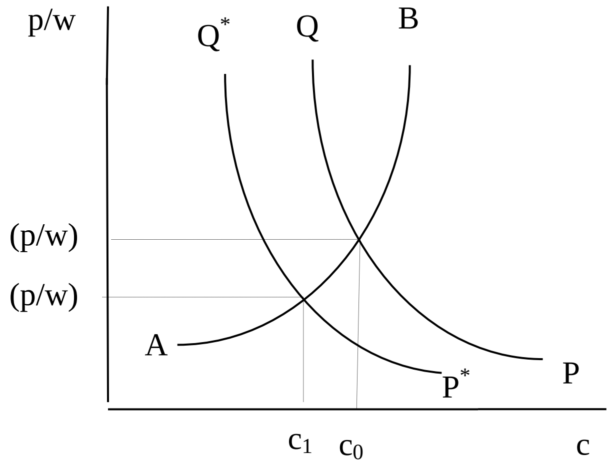


**Figure 3.2: Effect of trade on real wage and consumption**

The intersection of AB and PQ curves determine the equilibrium values of (p /w) and c. The equilibrium number of products (N) is determined as follows:

$$L = \sum_{i=1}^{N} L_i = \sum_{i=1}^{N} (\alpha + \beta y_i) = N(\alpha + \beta y_i) = N(\alpha + \beta Lc)$$

$$N = \frac{L}{(\alpha + \beta Lc)} = \frac{1}{(\alpha + \beta Lc)/L} = \frac{1}{\left[(\alpha / L) + \beta c\right]} \qquad (6)$$

where L is to total labour force in the country. The assumption of symmetry means that all firms are alike and hence we can simply replace the summation sign $\left(\sum_{i=1}^{N}\right)$ with N. The term y is replaced as $L_c$ as total output of each variety (y) is equal to the total demand for each variety ($L_c$). Therefore, given L and the fixed parameters α and β, the value of N is endogenously determined once the equilibrium value of c is known. Thus, equilibrium in the closed economy is fully determined.

What will happen when the home country `A' opens trade with the foreign country `B', which is identical with `A' in every way? In the H-O-S setting, there is no reason for these countries to engage in trade as firms in the two countries are symmetric (prices and quantities are identical across varieties in the two countries). But the story is different in the present model in that trade in differentiated commodities will take place even if identical ranges of varieties are produced in both countries in autarky. The reasoning for this goes as follows. In the absence of trade barriers, countries A and B together constitute an `integrated economy' (a term to represent two or more open economies combined), which is equivalent to twice the size of either of the two countries. Firms in this ‘integrated economy’ have no incentive to produce the same varieties as other firms because every variety enters into consumer's utility function symmetrically. Since no two firms produce the same variety, each variety will be produced in only one country. At the same time, all varieties are demanded in both the countries, which necessitate international trade in different varieties of the same commodity Y and hence IIT.

Two identical countries trading is just like doubling the population L, which will have no impact on AB curve. However, the PQ curve will shift down (see equations 4 and 5). As a result of this shift the following changes occur: (i) consumption of each variety falls from $c_0$ to $c_1$ ; (ii) p/w falls from $(p/w)_0$ to $(p/w)_1$ which means that real wage w/p rises; (iii) number of varieties (N) available to consumers in each country increases due to the rise in the total number of consumers (L in integrated economy) and fall in the consumption of each variety (see equation 6); (iv) output produced by each firm (y) increases[1]; and (v) number of varieties produced in each country falls. The last result means though the number of varieties available to consumers in each country go up, the number of varieties produced in each country falls. The number of varieties produced should necessarily decline in order to satisfy the employment condition in each country.

$$L = N(\alpha + \beta y)$$

$$N = \frac{L}{(\alpha + \beta y)} \qquad (7)$$

Given that productive resources in each country (L) remains constant, an increase in the value of y means that N should necessarily fall. While the number of varieties produced in each country falls, the sum of varieties from both countries under free trade exceeds the number in any single country before trade. Larger number of varieties for consumption means that consumers in each country clearly gains from trade. Higher real wages are another source of gains from trade in this model. The model shows that trade liberalization leads to a 'scale effect' (surviving firms expand their outputs) as well as a 'selection effect' (some firms are forced to exit).

Ideal variety approach developed by Lancaster (1980) and Helpman (1981) differ from Krugman's model in that the symmetry assumption of the utility function is not used. Instead, in this approach, every consumer residing in the country A is assumed to have preference for an ideal variety. Consumers differ in terms of the most preferred variety, but the assumption of equal density of preferences implies that the same aggregate demand exists for every variety. The supply side is characterised by free entry and exit, identical cost function for all varieties, and the existence of scale economies in the production of every variety. Because of the existence of scale economies, the number of varieties produced cannot be infinite suggesting that some consumers will be unable to get their ideal variety. Firms entering the market must decide on the particular variety and price. Profit maximising behaviour will ensure that no two firms will produce the same

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[1] This is evident from equation (5). Since p/w falls while α and β remain constant , it is clear that Lc = y should necessarily increase.

variety in equilibrium. Also, in equilibrium, each variety will be produced in the same quantity, will be sold at the same price and each firm will earn zero profit. This situation is referred to as perfect monopolistic competition.

Now, consider the possibility of trade with a foreign country B which is identical with `A' in every way. The symmetry assumption will ensure that the autarkic equilibrium features of A and B are identical and the `integrated economy' is equivalent to twice the size of either of the two countries. Also, in the 'integrated economy', each variety will be produced by only one firm and therefore in only one country. All varieties will, however, be consumed in both countries, giving rise to IIT.

The basic models outlined above have been extended by integrating them with the H-O-S approach to international trade. Generally, this has been done by introducing a second output sector (such as, agriculture, producing a homogenous good under conditions of constant returns) and cross-country differences in factor endowments. The integrated models predict that trade based on comparative advantage and trade based on scale economies will take place side by side.

## Check Your Progress 3

**Note:** i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) What is the effect of intra-industry trade on real wage and consumption?

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2) What will happen when the home country 'A' opens trade with the foreign country `B', which is identical to 'A'?

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## 3.3.3. IIT in Vertically Differentiated Commodities

The horizontal models are considered to be of greater relevance for understanding the occurrence of bilateral IIT among developed countries. A different approach pertaining to IIT between countries with clearly different factor endowments takes its point of departure in vertical product differentiation. Most economists formulated models of vertical IIT without recourse to economies of scale. In general, these models predict the pattern of IIT along the lines similar to the pattern of inter-industry trade predicted in the conventional trade model, according the central role to factor endowment differences. What follows is a brief discussion of the essential elements of the vertical models.

Falvey (1981) adopted a partial equilibrium approach by concentrating on trade within a single industry, which is assumed to posses a stock of industry specific capital and produces a continuum of products differentiated by quality. Higher quality products are characterised by higher capital labour ratio used in their production. Therefore, the comparative advantages of capital abundant countries lie on the higher ends of the quality spectrum and that of labour abundant countries lie on the lower ends. Vertical IIT between two countries will arise given an overlap in the demand for different qualities and this possibility is enhanced the greater is the factor endowment difference between the countries. Assuming that relative capital abundance is reflected in relative income per capita, the model suggests a hypothesis contrary to what emerges from the horizontal models. That is, the intensity of IIT will be positively correlated with the per capita income differences between the trading partners. Another testable hypothesis, which emerges from the model, is that the shares of IIT vary inversely with the levels of trade restriction of partner countries.

The supply side in the Falvey and Kierzkowski (1987) model is a close descendant of Falvey (1981), the demand side being fully elaborated in the former. Each individual is assumed to demand only one type of differentiated product and given relative prices, this preferred quality is determined uniquely by the individual's income. Since the aggregate income of an economy is not equally distributed, there is, at any point in time, an aggregate demand for a variety of differentiated goods. This holds well in the `integrated economy' as well and consequently vertical IIT will emerge, the intensity of which being higher the more dissimilar (in income distribution) are the trading countries. The model also suggests that the shares of IIT will be positively correlated with the market size of countries.

In the preceding models, capital assumes the decisive role among the factors of production in determining the product specification. Whereas, Flam and Helpman (1987), in their model, ascribe the central role to labour. According to them, population in every country is characterised by a non-degenerate distribution of skills and differences in skills are reflected in differences in the endowment of effective labour supply. Higher quality products are characterised by relatively larger inputs of labour used in their production and the pattern of IIT reflects differences in technology and in income distribution.

A strikingly different mode of analysis focuses on the markets where R&D expenditure (considered a sunk cost) is a prerequisite to quality improvement. Given that the average variable cost rises only slowly with quality, the number of firms in the market is bounded; by extending the distribution of income the upper bound on the number of firms increases, however. Thus, the `integrated economy' can support a large number of firms if the two countries are more dissimilar in income distribution. The higher income country specialises in higher quality products and the lower income country specialises in lower quality products resulting in vertical IIT.

### 3.3.4. IIT in Intermediate Products

Ethier (1979, 1982) focuses on the significance of three important aspects to international trade: international returns to scale, product differentiation in producer goods and interdependence of world industrial activity. He points out that scale economies resulting from an increased division of labour depend on the size of the world market rather than the production level in any one country. The interdependence of world industrial activity implies global dispersion of the distinct operations consisting of production process in an industry, which in turn depends on the efficiency gains from increased specialisation.

The analysis produces a model of IIT in intermediate products, wherein the desire for product variety arises as a consequence of larger markets permitting greater division of labour. This is in contrast with the models discussed earlier wherein the desire for product variety arises from consumer preference. The analysis further demonstrates that internal scale economies and product differentiation are not the only factors determining the degree of intra-industry trade.

In agreement with the horizontal models, Ethier's model also predicts that similarities of factor endowments between nations tend to promote IIT. Also, if differentiated intermediate goods are tradable, the analysis of trade pattern is similar to that of differentiated final goods.

### 3.3.5. IIT in Identical Commodities

The usual approach to IIT is to assume that such trade arises because slightly different commodities are produced and traded to satisfy the desire for variety. However, Brander (1981) and Brander and Krugman (1983) have shown that there are reasons to expect IIT even in identical products. In their models, the pattern of trade is determined by the interaction of increasing returns to scale, transport costs, and oligopolistic behaviour of international firms. A Cournot strategy is assumed in which each firm maximises profit assuming the output of other firms in each market (domestic and foreign) remains the same. The market equilibrium involves IIT despite the fact that both countries A and B (symmetric) produce identical commodities, and there is an obvious loss due to transport costs. (Transport costs, however, should not be too high. As transport costs fall, the proportion of foreign goods in domestic consumption increases.) IIT is the result of what is called “reciprocal dumping"(also called "cross-hauling"), i.e., each firm dumps into other firm's home market. The term "dumping" is because the f.o.b price of export is less than the domestic price. The firm's incentive to bear the higher effective marginal cost (including transport costs) associated with exporting stems from the possibility of its marginal revenue in the foreign market exceeding that in the domestic market because the market share of each firm in the foreign country is smaller than that in the home country. This is an equilibrium situation because imperfectly competitive firms set marginal revenue equal to marginal costs in different markets. The drawback of this model is that the Cournot assumption ignores strategic interdependence of firms usually observed in oligopolistic markets. In particular, the ability of this model to explain the pattern of IIT in the context of developing countries is suspect.

## Check Your Progress 4

**Note:** i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Describe Intra-Industry trade in Intermediate goods.

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2) What could be the possible reasons behind IIT in identical products?

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3) Discuss the relevance of vertical IIT.

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## 3.4 MEASUREMENT OF IIT

Two major issues in the empirical analysis of IIT are concerned with: (i) the definition of ‘industry’ and (ii) measurement of IIT. The measurement issues were subjected to controversies ever since they were first raised and now there exists some professional consensus on most of the issues.

### 3.4.1 Defining `Industry’ and the Problem of Categorical Aggregation

Operationalizing the concept of ‘industry’ is vital for the empirical analysis of IIT. It is, however, easier said than done, as there is no agreed single definition of ‘industry’. Mainly, two criteria have been followed in economic analysis. The first considers different products as belonging to a single industry if they are good substitutes in production, while the second criterion is related to the extent of substitutability of products in consumption.

A major issue for the empirical analysis is that the criteria for official trade classification do not coincide exactly with the criteria for products to constitute IIT. The problem of categorical aggregation arises as a result of the inclusion in a single trade category of products, which either have different production functions or have different end uses. Importantly, the extent of IIT, as estimated by the usual measures is sensitive to the level of aggregation (of products) which defines an industry: higher levels of aggregation usually give rise to greater extent of recorded IIT. The use of a more detailed classification system for the measurement of IIT does not necessarily solve the problem because it would tend to separate commodities that are good substitutes in production and consumption.

Thus, the degree and kind of homogeneity of commodity items under each trade category need to be considered for the purpose of empirical analysis. The most common practice is to select a specific level of statistical aggregation in the official classification as the best approximation to the concept of industry. Now, there exists a professional consensus to consider the products grouped under 4 or 6-digit level of Harmonized System (HS) of Classification as roughly corresponding with the concept of an industry. In general, commodities grouped under a particular 4 or 6-digit code may be close substitutes in production, consumption or both. First, there may be products traded having similar input requirements, are close substitutes in production, but not in consumption. Secondly, there are products traded that are close substitutes in consumption, but not in production. A third category involves very similar goods, which are close substitutes in consumption and production. Finally, the exchange of absolutely identical products can also be found. In practice, the simultaneous exchange in these four types of commodities is considered as IIT, while other types of transactions constitute inter industry trade.

### 3.4.2 Measuring the Intensity of IIT

The intensity of IIT is usually measured by using the well-known Grubel Lloyd (GL) index:

$$GL_i = \frac{(X_i + M_i) - |X_i - M_i|}{(X_i + M_i)} \times 100, \qquad (8)$$

where $GL_i$ is the IIT index for industry $i$, and $X_i$ and $M_i$ are values of exports and imports in industry $i$. The value of $GL_i$ ranges from 0 to 100. If there is no IIT (i.e., one of $X_i$ or $M_i$ is zero) $GL_i$ takes a value 0. If all trade is IIT (i.e., $X_i = M_i$), $GL_i$ takes a value of 100. Grubel and Lloyd (1975) proposed the following weighted index to arrive at an aggregate measure of IIT:

$$GL = \frac{\sum[(X_i + M_i) - |X_i - M_i|]}{\sum(X_i + M_i)} \times 100. \qquad (9)$$

The following features of Grubel Lloyd index are worth noting.

1) $GL_i$ is independent of the absolute value of exports and imports, thus it does not allow conclusion on the absolute value of IIT.

2) $GL_i$ is symmetric in the rise in exports and imports, thus it could rise due to a rise in either exports or imports or both.

3) $GL_i$ is non-linear in changes, so a small alteration in the coefficient would disguise a big change in trade volumes and vice versa.

4) $GL_i$ is subject to aggregation problems. Grubel and Lloyd argue that "…the measure of IIT at a more aggregative level is greater than, or at least no less than, the measured intra-industry trade with a finer commodity breakdown".

5) Grubel and Lloyd observed that GL is a biased downward measure of IIT if the country's total commodity trade is not balanced or if the mean is an average of some subset of all industries for which exports are not equal to imports. They considered this an undesirable feature of a measure of average IIT and suggested the need for correcting for trade imbalance. They proposed to use an adjusted measure. Although, many economists disagree with the idea of adopting adjusted measure.

### Check Your Progress 5

**Note:** i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Describe Grubel Lloyd (GL) index for Measuring the Intensity of IIT.

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2) What were the major issues in the empirical analysis of IIT?

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## 3.5 LET US SUM UP

In this unit, we discussed about Intra industry trade and how trade liberalization has supported its growth. IIT means the simultaneous export and import of products, which are very close substitutes for each other in terms of factor inputs and consumption. Growth of IIT is an outcome of greater specialisation in the manufacturing of unique varieties or product lines by the individual plants in different countries. In a liberalized environment, intensity of IIT has increased in most of the countries. In absence of trade barriers, IIT has increased in many industries.

We also distinguished between the two types of IIT – horizontal and vertical IIT. Horizontal IIT refers to the simultaneous export and import of the differentiated products within an industry which are of a similar quality. Vertical IIT, on the other hand, refers to the exchange of differentiated products that are of different qualities. While discussing the theories of Intra-Industry Trade, we also discussed about the need for separate theory for Intra-Industry Trade. We discussed about 'love for variety' and 'ideal variety approach' in case of horizontal IIT. We also discussed about IIT in identical goods and intermediate goods. We further discussed about the major issues in the empirical analysis of IIT and Grubel Lloyd (GL) index for measuring the Intensity of IIT.

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## 3.7 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

### Check Your Progress 1

1) Refer to Section 3.2
2) Horizontal IIT refers to the simultaneous export and import of the differentiated products within an industry which are of a similar quality. Vertical IIT, on the other hand, refers to the exchange of differentiated products that are of different qualities.
3) Refer to Section 3.2

### Check Your Progress 2

1) Refer to Sections 3.3.1
2) "Love-of variety approach" considers that each consumer has a taste for many different varieties.
3) "Ideal variety approach", which considers that each consumer has a desire for the most preferred variety rather than many varieties.

### Check Your Progress 3

1) Refer to Section 3.3.2 and Fig. 3.1
2) In the 'integrated economy', each variety will be produced by only one firm and therefore in only one country. All varieties will, however, be consumed in both countries, giving rise to IIT.

### Check Your Progress 4

1) Refer to Section 3.3.4
2) Refer to Section 3.3.5
3) Refer to Section 3.3.3

### Check Your Progress 5

1) The Grubel Lloyd (GL) index is used to measure the intensity of IIT.
2) Refer to Section 3.4.1

# UNIT 4 ALTERNATIVE EXPLANATIONS OF TRADE *

## Structure



## 4.0 OBJECTIVES

After studying this Unit, you should be able to:

- explain new, complementary trade theories based on real world situations using concept of Economies of Scale, Imperfect Competition, Product differentiation, Technological Gap etc.;
- examine the effect of relaxing each of the classical and Heckscher-Ohlin trade theory assumptions; and
- critically examine the basics for real world international trade patterns in world market.

## 4.1 INTRODUCTION

The main objective of this unit is to discuss new and complementary trade theories to convey the current patterns of international trade that are closer to reality with less restrictive assumptions. Absolute Advantage, Comparative Advantage and Heckscher & Ohlin theories left unexplained a substantial share of today's international trade. Many theories have been developed after World

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War II to explain Trade and international production. These theories find out a number of factors that could explain International Trade flows between different countries, linking the aspect like cost reduction and economies of scale. Thus, this current unit explains complementary theories of international trade on the basis of economies of scale, imperfect competition and differences in technological changes between countries.

Classical trade theories explained the pattern of international trade using absolute advantage, comparative advantage and on differences in factor endowments between countries. However, these theories could not explain significant pattern of today's international trade. Relaxing the assumption of perfect competition, constant economies of scale, and no differences in technological changes between countries however required new empirical theories to provide substantial evidence of current pattern of international trade, which was not explained by classical trade theories.

The assumption of Heckscher-Ohlin theory is not generally valid as these theories assume both nations use the same technology in production. In the reality, most of the countries are using different technologies for the production. Technological gap model and product life cycle models explain these phenomena. Both of these models are viewed as dynamic extensions of the basic Heckscher-Ohlin model. These models are discussed in section 4.2.

Classical trade theories assumed constant return to scale, but international trade could be also based on increasing return to scale. According to current trade pattern in international trade, though two countries are similar in every aspect there is still a basis for mutually beneficial trade because of economies of scale. According to concept of economies of scale, when each country specializes in the production of one product, then the world total output is greater than without specialization. This was unexplained in earlier classical trade theories; (see Unit 1) hence section 4.3 will explain the concept of increasing return to scale and economies of scale to validate recent trade flows between countries.

Another significant development in recent years in international trade flows are a large portion of international trade today involves of exchange of differentiated products which the intra-industry trade model in current chapter under section 4.4 explains. Comparative advantage theory fails to explain the intra-industry trade that arise to take advantage of important economies of scale in the production in different countries.

Recently most of trade flows between countries are also affected by distance between two destinations. Gravity Model explains this concept in this chapter under section 4.5. The main focus of this model is to forecast bilateral trade flows between countries based on the distance within two countries and their respective economic dimensions. Gravity model is now used more frequently to empirically analyse trade patterns by using distance between two countries, taking the help of physical distance, cultural distance, etc.

Another important current phenomenon can be viewed from the angle of transportation and logistics costs in the international trade. Logistics cost reduce the volume and benefits of international trade. A classical theory does not consider transportation or logistics cost in the explanation of international trade. An environmental standard also affects the location of industry and international trade which is explained in this chapter at the end in section 4.7.

Krugman also significantly explained his alternative international trade theory by considering internal returns to scale and adopting a recent innovation model under monopolistic competition. Krugman's alternative theory of international trade based on more realistic assumption is explained in this chapter under section 4.6.

Relaxing the assumption of classical trade theories only modifies the concepts but does not invalidate the classical trade theories. These theories are only the extension of classical trade theories with more realistic assumptions which explain the current patterns of international trade in the world market.

## 4.2 TECHNOLOGICAL GAP MODEL AND PRODUCT LIFE CYCLE THEORY

According to the technological gap model introduced by Posner in 1961, most of the trade in industrialized countries is based on introducing new products and new production processes. Introduction of new product and process creates temporary monopoly for innovative firm and nation in the world market by protecting their products by patents and copyrights. The main objective of this protection is to encourage flow of inventions in the market.

According to technological gap model, technologically advanced countries export technology intensive goods. However, by technological know-how, foreigners or underdeveloped and emerging markets also adopt the foreign technology and again export their product to high technological countries with some value addition. This kind of trade occurs as labour cost is lower in the underdeveloped or in emerging markets which creates competitive advantage to emerging markets. But this model fails to explain the reasons, size of technological gaps and how technological gap is eliminated over time.

Product life cycle model explains a simplification and addition of technological gap model. The main objective of Product life of cycle theory is the improvement of trade theory beyond David Ricardo's static framework of comparative advantages. Raymond Vernon explained this theory in 1966, where he discussed three stages of production for various manufactured goods. Vernon focused on the dynamics of comparative advantage and depicted inspiration from the product life cycle to explicate how trade patterns change over time. This theory tries to explain an internationalisation process of a firm where a local manufacturer in a developed country begins selling a new, technologically advanced product to foreign markets. As demand from international markets rises, production gradually shifts overseas enabling the firm to maximise economies of scale and to avoid trade barriers, generally known as tariff jumping hypothesis[1]. In the later stage when product matures and becomes more of a commodity, the number of competitors increases, hence production shift to lower cost countries or labour-intensive countries.

He argued that comparative advantage for many products may shift over time from one country to another because these products go through a product life cycle. The main assumption of this theory is that most manufactured products follow a life cycle in which these products first appear as innovations and eventually become completely standardized. First stage of the production is called 'innovation' where the new product is introduced in to the market. In this stage, manufacturers mainly concentrate on the degree of freedom of their product and mainly invest in the product's research and Development (R &D). In the first stage the production, product is designed, developed and produced locally in the home market in limited number. Subsequently, after the establishment of the product in the home market, the foreign demand is mostly met by undertaking export.

In the second stage the production, manufactured goods became 'mature'. In the second stage of production, the price elasticity of demand for the product is comparatively low. The demand for the product increases in the foreign market and competitors appear in the international market. At this stage, producers begin manufacturing in the foreign country to supply products in the foreign market and try to decline costs with international production to compete with the competitors. Therefore, in the second stage the firm becomes international.

In the final stage product become standardized. The production technique becomes well known and reaches to the most of the competitors. Firms moved their production to lower cost and lower wage countries. The product is exported to the original country of innovation where the product is phased out in order to favour innovation of new product. Thus, the exporter becomes importer in this stage of production and national market will have to import relatively capital intensive products from low income countries.

According to product life cycle theory, when a new product is introduced in the market, its production process requisite highly skilled labour but as product matures or become standardized then production process required less skilled labour. Therefore, comparative advantage shifts from developed countries to less developed countries. The main example of product life cycle theory is the evaluation of Xerox photo-copier manufacturing plants which started originally in the USA in the 1940s and 1950s. In the first stage of production Xerox

[1]This hypothesis states that MNCs (Multinational Corporations) attempt to jump over tariff or non-tariff barriers by establishing foreign subsidiaries.

exported its products then it shifted its manufacturing plant first to Europe and finally, as the technology matured, it shifted its manufacturing plant in India in the year 1970s and 1980s.

Product life cycle theory does not apply in all conditions; mainly products with small life span may not go through the life cycle described in this theory. Secondly, this theory does not hold true in the cases of luxury goods, whose perceived value mainly depend on marketing campaigns.

## 4.3 ECONOMICS OF SCALE AND INTERNATIONAL TRADE

One of the significant incentives for international trade is the productivity enhancements that can arise because of economies of scale in production. After 1980's trade economists introduced the concept of economies of scale under new trade theory. One of the main assumptions of the Heckscher-Ohlin model was that products are produced under condition of constant return to scale, but the new trade theory assumes increasing return to scale, where output grows more than the increase in inputs or factors of production.

According to Economies of scale concept, the per-unit cost of producing a product falls as the scale of production rises holding input prices as constant. Therefore, larger production reflects a lower cost due to the specialization e.g., China make production at larger scale which lowers down production cost and prices of many goods. As a result of lower cost, prices of the commodities get lower and exported to other countries. Economies of scale, specialization and trade can enhance in the world productivity and welfare benefits that accrue to all trading partners.

The main assumption of this concept is under economies of scale, trade between two countries does not depend on country differences. Indeed, it is likely that countries could be equal in all respects and yet find it advantageous to trade. Thus, economies-of-scale models are frequently used to explicate trade among countries like the USA, Japan, and the European Union. Both the countries and EU and other advanced countries have similar technologies, similar endowments, and to some extent similar preferences. Expending classical models of trade, these countries would have little reason to involve in trade. However, trade between the advanced countries makes up a significant share of world trade. Economies of scale can provide an answer for this type of trade.

The idea of economies of scale can be explained by figure 4.1. Let us assume that two countries India and China are similar in every aspect and producing two commodities, wheat and cloth. In the absence of Trade India produces and consume at point A and China at point $A_1$. With trade, India could specialize completely in the production of commodity wheat as India has comparative advantage in the production of wheat. Similarly, China can specialize in the production of Cloth as China has comparative advantage in the production of

Cloth. By exchanging 110 quantity of wheat to 70 quantities of cloth with each other, India ends up consuming at E, and gains 20 quantities of wheat and 10 quantities of Cloth, while the China ends up consuming at $E^1$ and gains 30 quantities of Wheat and 10 quantities of Cloth. Thus, both the countries get extra quantities of commodities when they specialize in the production as it creates economies of scale in the production.

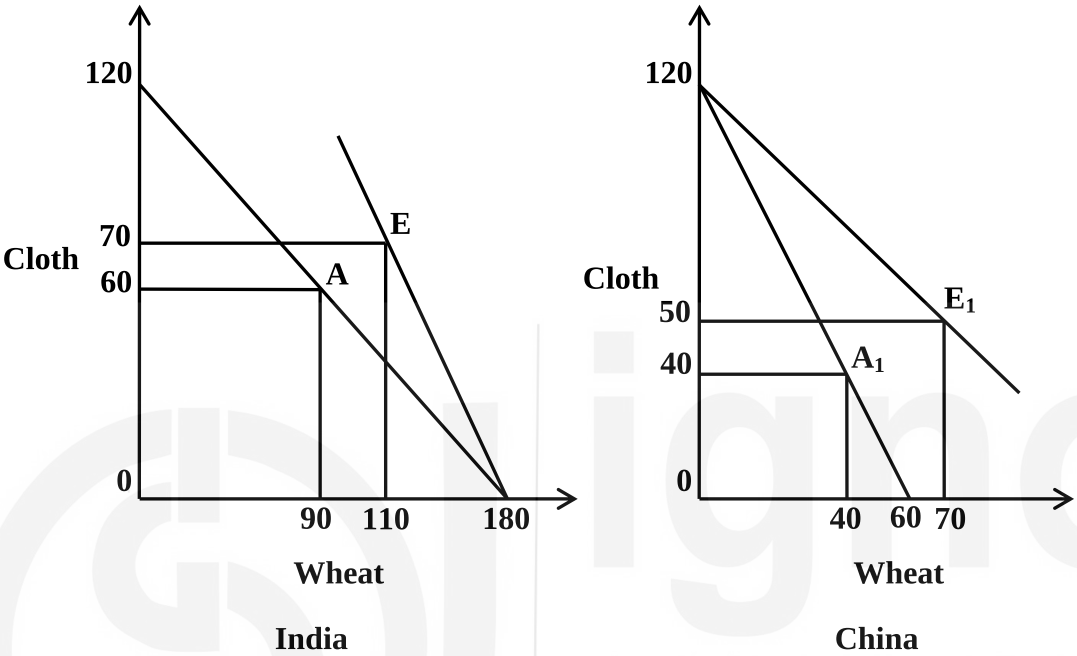


**Figure 4.1: Trade based on Economies of Scale**

Linder in 1961 developed an advanced hypothesis similar to economies of scale where he explained nation exports those manufactured goods for which a big domestic market exists. As there is more demand in domestic market, in the process of satisfying such a market, country obtains the necessary experience and efficiency. This experience and efficiency in manufacturing of these products further converts into increasing export of these commodities to other countries that have similar tastes, preferences and income level. The nation will import those commodities that appeal to its low and high income minorities. According to this overlapping demand hypothesis, trade in manufactures is likely to be largest among countries with similar tastes and income levels.

## Check Your Progress 1

**Note:** i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) What is meant by technological gap model? Discuss the pattern of international trade according to technological gap model.

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2) Discuss the concept of product life cycle model. What are the various stages in product life cycle?

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3) Discuss the concept of economies of scale? How can economies of scale define current pattern of international trade in recent years?

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## 4.4 PRODUCT DIFFERENTIATION AND INTERNATIONAL TRADE

Currently, most of the international trade is based on differentiated product rather than homogeneous product in world market. For example, Audi is not identical to a Hyundai, Land Rover, Honda or a BMW. Therefore, most of the international trade occurred within differentiated products of the same industry or broad product group in same HS code[2] known as intra industry trade as opposed to inter-industry trade in completely different products. We have already discussed about Intra-industry trade in details in Unit 3. Let us discuss briefly here as a part of alternative explanations of trade theories. Intra- industry trade models are developed by various trade economists like Helpman, Krugman, Lancaster and others since 1979. Intra-industry trade or trade of similar products remains unexplained in the classical and Heckscher-Ohlin international trade models and theories. Comparative advantage theory earlier only explained the pattern of inter-industry trade. Intra- industry trade between countries occurred due to economies of scale in production. Competition within market forces each firm, plant or industry to focus on only few or one variety and styles of the same product rather than several diversified varieties and styles. This strategy is adopted by many firms to keep unit cost low as specialization decreases per unit cost of production. Thus, country can import the other varieties and styles from another country. Intra- industry trade lead to overall welfare of consumer as consumer received wider range of products at lower prices due to economies of scale in production. This can be explained with the example of European Union (EU). Before formation of EU, plant size of United States (US) firms and European firms was similar, but the unit cost were much higher in Europe due to many more varieties and styles of a product compared to US firm. When EU

[2]The Harmonized Commodity Description and Coding System, also known as the Harmonized System (HS) of tariff nomenclature is an internationally standardized system of names and numbers to classify traded products. It came into effect in 1988 and has since been developed and maintained by the World Customs Organization (WCO) (formerly the Customs Co-operation Council), an independent intergovernmental organization based in Brussels, Belgium, with over 200 member countries

formed and tariffs were reduced each plant specialized in the production of only few goods which sharply declined per unit cost in European Union.

The main reason for inter-industry trade is when we aggregate trade data at 2 digit level or HS-2 level; many products are aggregated into one category. For example, India imports and exports similar products like petroleum products, automobiles and so on. Petroleum products like crude oil, petrol, and diesel are aggregated into one category of product code under HS-2 code but actually India imports crude oil from Middle East countries which is processed in coastal areas of India and exported back to another country. Similarly, India imports components of automobiles from Japan as Japan has a comparative advantage in manufacturing of auto-components. These auto-components are assembled in India and exported back to another country. Most of the assembly lines and crude oil refineries are in India as labour is cheaper in India or India is labour-intensive country.

However, since all these types are generally aggregated into one export or import category, it could appear that the countries are exporting and importing "identical" products when they are actually exporting one type of automobile or petroleum products and importing another type. When there are no differences in resource endowments or technologies between countries' economies of scale, discriminated products can explain intra-industry trade through is known as monopolistic competition model. Monopolistic competition model emphasise on consumer demand for a multiplicity of characteristics embodied in the goods sold in a same HS code category. In this model, trade is possible even when countries are very comparable in their productive capabilities.

The level of intra-industry trade can be measured by the intra- industry trade index by following formula:

$$T = 1 - \frac{|X - M|}{X + M}$$

Where T represent intra-industry trade index, X represent value of exports and M represents value of imports of a particular industry or commodity group and the vertical bars in the numerator of equation denote the absolute value. Intra industry trade index (T) value ranges from 0 to 1. T = 0 represent that a country only exports or imports the good or there is no intra industry trade and T=1 represent that intra- industry trade is maximum or exports and imports of a goods are equal. T index is calculated by various empirical research studies and found that current most of the trade is guided by intra- industry trade. In 1967, Grubel and Lloyd calculated T index for 10 industrial countries and found that combined weighted average of T for all industries in all 10 countries was 0.48. This result indicates that in 1967 nearly half of the trade among 10 industrial countries involved the exchange of differentiated products of the same industry.

## 4.5 GRAVITY MODEL OF TRADE

The Gravity Model of Trade is a key and robust empirical model in the international trade used to understand the pattern of international trade. This model was firstly introduced and used by Jan Tinbergen in 1962 employing gravity equation which is derived from Newton's theory of gravitation. As globes are attracted to each other in terms of their sizes and proximity similarly countries are also attracted by their sizes and proximities. According to gravity model, bilateral trade between two countries is proportional to size, measured by GDP, and inversely proportional to the geographic distance between them. The main focus of this model is to forecasts on the bilateral trade flows and these forecasts are based on the distance within two countries as well as their respective economic dimensions.

It is important to note that, the gravity model was not based on any theory but most of the trade economists empirically derived the impact of size and distance of the country on bilateral trade flows. As most of the trade economists used gravity model to explain pattern of trade therefore this model has been seen as a significant development on previous trade theories of Ricardo and Heckscher-Ohlin. Before introduction of gravity model none of the theoretical trade model used the size of an economy significantly in their explanation.

This model is important in terms of forecasting the impact of changes in trade policy on trade costs. This model is also flexible in terms of distance where it can cover different variables like tariff imposed at a border, cultural, institutional and political differences between trading nations. The Gravity model has recently used for understand the impact of BREXIT on trade flows between UK, EU and other countries.

Moreover, the gravity model is no longer concerned only with trade in goods, but has recently been applied with success to trade in services.

**Intuitive gravity model:**

Firstly gravity model was introduced as an intuitive way of understanding trade flows between countries. In its basic form, the gravity model can be written as follows:

$$log\, X_{ij} = c + b_1\, log\, GDP_i + b_2 log\, GDP_j + b_3\, \log\tau_{ij} + e_{ij} \ldots\ldots\ldots\ldots\ldots\ldots\ldots (1a)$$

$$\log\tau_{ij} = \log\,(\text{distance}_{ij})\text{------------------------------------------}(1b)$$

In this model $X_{ij}$ represent exports from country i to country j, GDP represent each country's gross domestic product, $\tau_{ij}$ indicates trade costs between the two countries, distance represent the geographical distance between two counties, as an observable proxy for trade costs and $e_{ij}$ is a random error term. The c term is a regression constant, and the b terms are coefficients to be estimated.

The term “gravity” originates from the fact that the nonlinear form of equation 1a look like Newton’s law of gravity: bilateral trade between two countries is proportional to size, measured by GDP, and inversely proportional to the geographic distance between them. Putting into another words, gravity model explain that we expect similar countries trade more, but we assume countries wither larger distance trade less, possibly because transport costs between them are higher.

However, the intuitive gravity model has some limitation. For example, If India enters into a Free Trade Agreement with EU that lowers tariffs on their respective goods. Basic trade theory indicates that it will be also affect the trade with USA even though it is not party to the agreement. One of the important concepts of trade creation and trade diversion are examples of such effects. But intuitive gravity model does not account the impact of preferential trade agreement or free trade agreement on third country. This is clear from equation 1a which indicates that, $\frac{dlogx_{ij}}{dlogt_{ik}} = 0$

Therefore, decreasing trade costs on one bilateral route does not affect trade on other routes in the basic model, which contradicts standard trade theory.

A second problem with intuitive gravity model rises if we consider equal reductions in trade costs across all routes, including internal trade. For example there is decline in the price of crude oil, which drops transport costs everywhere, including inside countries. In the intuitive gravity model, this change would result in proportionate increases in trade across all bilateral routes, including internal trade. However, such an outcome is not considered in this model despite the change in trade costs as relative prices have not changed at all. In the absence of a change in relative prices, economic theory assumes that consumption patterns to remain constant for a given amount of total production measured by Gross Domestic Product.

As discussed in the previous sections there are problem with intuitive model of gravity hence this model was alerted by some trade economist. The first attempt in this regard was made by Anderson in 1979. However, the applied literature considered theoretical gravity model after the famous “gravity with gravitas” model of Anderson and Van Wincoop in 2003. This model is essentially a demand function which assumes that utility of consumers increases from consuming more variety of goods. From the production side, this model assumes that firm produces a single, unique product variety under increasing return to scale.

## Check Your Progress 2

**Note:** i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Critically discuss the concept of Intra-Industry Trade. How international trade is based on product differentiation?

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2) What is the Gravity model of trade? Explain with the help of formula and examples.

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## 4.6 KRUGMAN ALTERNATIVE THEORY OF TRADE

Krugman explained his theory of international trade by considering internal returns to scale and adopting a recent model of innovation under monopolistic competition. Internal economies of scale suggest that per unit average output cost will decrease as the total output increases. Most of the products have high fixed costs, and as production increases, it spreads the fixed cost of product. According to Krugman, if there are internal economies of scale, markets has fewer firms and each firm will produce more and discriminate their product from their competitors. Total number of firm is determined by average cost and price. When the price is higher than average, new firms will enter the market. Krugman also explained that larger economies have higher demand, hence these economies has higher internal return to scale.

Internal returns to scale is also applicable to International trade when there is more trade between countries. Suppose if trade between India and EU suddenly emerged due to free trade agreement then production in both markets will grow. There will be fewer total firms, and each persisting firm would produce more. Consumers in both countries would be able to buy variety of the products with low price. Per unit cost would also fall, because of the exploitation of further internal economies of scale. Therefore, even if there is no absolute or comparative advantage in trade, both countries will undertake trade to exploit the benefits of internal economies of scale. Countries will, all else same export the goods where market size or aggregate demand is highest or start their production process in a country where market size is higher by undertaking foreign direct investment. The main reason for this is that firms will be able to exploit higher internal economies of scale than other countries in that market. Therefore, according to Krugman countries will tend to export the product in which they have specialization or they are able to produce more under the concept of internal economies of scale.

When tariff and non-tariff barriers to trade are more, factors of production will tend to move to countries where there are economies of scale in production. For example, suppose labour is only one factor of production, trade barriers would induce labour to move to the country with largest market as real wages will be more and there will be extra product diversity in that market. Krugman also explained his idea considering trade cost in the form of transaction costs. Under transaction costs specialization will be more limited, as trade costs decline the profitability or exporting. Severities of Internal economies of scale also decide the degree of specialization. Country will be specializing more when internal economies of scale will be more for products. Therefore, higher trade cost and limits of internal economies of scale lead to intra-industry trade.

According to Krugman trade between countries is also dependent on difference in factor endowments. Nation will export the good which are relatively abundant. But if two countries are equally abundant, it will lead to intra-industry trade between countries. If the countries are equally abundant, the effects for changes in the distribution of income as a result of trade then trade will be Pareto optimal. The gains from the trade and income distribution depend on the severity of difference between factor endowments. In case of more unique country's factor endowment, the more relatively scarce factor will lose from trade and the relatively abundant factor will gain. The international market scare factor will lose from the trade because of the lower price of that product in that country as it faces competition from foreign producers who have lower costs because of relative abundance. Whether trade is Pareto optimal is determined by whether the benefit increase from product difference is great enough to make up for the comparative loss of income for the scarce factor.

Comparative advantage theory fail to explain trade based on internal economies of scale therefore Krugman explained the idea of internal economies of scale and monopolistic competition. If there are no product differences in relative costs, technology and tastes, gains from trade can be generated from lower prices and greater product diversity. This is an alternative approach to the division of labour and trade theory.

## 4.7 COST OF LOGISTICS, ENVIRONMENTAL STANDARDS, AND INTERNATIONAL TRADE

Classical trade theories assumed that there is no transportation cost or transportation cost is zero, but in actual case significant cost is needed for transporting good from one place to another place in international trade. Cost of transportation or logistics cost includes freight charges, warehousing costs, costs of loading and unloading, insurance premiums, and interest charges while goods are in transit. Trade between two countries will occur only if the pre-trade price difference in the two countries exceeds the cost of transporting the good from one country to another country. Most of the commodities are not traded internationally due to high logistics costs. For example, Cement is not transacted internationally due to high transportation cost because of its high weight to value ratio. Cement is only traded to border areas due to high transportation cost.

Generally price of non-traded goods and services are determined by domestic demand and supply, while price of traded goods and services are determined by demand and supply in the world market. Currently, logistics cost between countries are significantly declined which resulted in increase in international trade. For example, Fish, Meat, fruits and vegetable found in many European countries in winters are shipped from country like India. In the past, due to high transportation cost and unavailability of refrigerator containers many perishable goods used to spoil in the transportation phase. Logistics costs are much higher in developing countries rather than developed countries and denote a significant obstacle to international trade for industries in developing countries.

Transportation cost can be analysed by two methods. One is by general equilibrium model and another by partial equilibrium analysis. Partial equilibrium analysis assumes constant exchange rate and constant level of income in two countries.

Logistics cost also influence the location of production under the international trade. Industries can be classified as market seeking, resource seeking or footloose industries. Resource seeking industries are establishing their production near to the source of raw material used by the industry. The main reason for this as the transporting raw material used by the industry is substantially higher than for shipping the finished product to market. For example basic chemicals, aluminium and steel which processes heavy and bulky raw material into lighter finished products. On the other hand, market seeking industries established there plant near to the markets. Market seeking industries normally produce goods that become bulky to transport during the production process. For example, cold-drink industries like Coca-Cola, Pepsi established there manufacturing plant near to market as bottling weight is higher than their raw material like concentrated syrup. Finally footloose industries are producing the goods that face neither substantial weight gain or lose during the production process. These industries tend to established near to availability of cheaper input cost. An example is provided by Japanese automobile industries which ship automobile machinery to India to be assembled due to cheap labour availability in India. Currently most of the countries are giving preferential tax treatment to attract these footloose industries.

International trade is also affected by environmental standards in different countries. Environmental standards refer to levels of air pollution, water pollution, thermal pollution and pollution resulting from garbage disposal that a nation allows. International trade which creates negative impact on environment can lead to serious trade problems because the price of traded goods and services often does not reflect the negative externalities or socio environmental costs. Countries with lower environmental standards are able to attract more polluting industries. For example, United States labour opposed NAFTA due to fear that it would lead to job losses as industries might move to Mexico to take benefit of more relax or less environmental laws compare to U.S market. According to world bank study in 1992, exports of polluting industries expanded faster than clean industries in the last two decades in the world market and mostly shifted to underdeveloped and developing countries from developed nation (Low, 1992). As country make progress, people demand more clean industries rather than polluting industries. Most of the developed countries are now taking help of environmental standards to protect their economy from polluting industries which also increased non-tariff barriers amongst the countries.

## Check Your Progress 3

**Note:** i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Discuss the Krugman's alternative theory of international trade.

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2) Discuss the impact of environmental standards and logistics cost on location of the production and international trade.

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3) What is meant by resource- seeking industries? Market seeking industries? Footloose industries? What determine the classification of the industry? How does this affect the pattern of international trade?

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## 4.8 LET US SUM UP

In this unit, we discussed about alternative explanation of international trade theories. Classical theories of International Trade unexplained significant portion of real International trade flows between countries. The current unit fill this gap by providing alternative explanation of international trade which is based on empirical evidences of current trade flow with more relaxed and realistic assumptions to explain current patterns of international trade. Alternative explanation of trade theories are based on concept of increasing return to scale, economies of scale, product life cycle theory, Gravity Model, Product differentiation. The current chapter ends with impact of cost of Logistics, Environmental Standards on International Trade and Krugman's alternative theory of trade.

## 4.9 KEY WORDS

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| **Increasing returns of scale** | An increasing returns to scale arises when the output increases by a larger proportion than the increase in inputs during the production process |
| **Economies of Scale** | According to Economies of scale concept, the per-unit cost of producing a product falls as the scale of production rises holding input prices as constant. |
| **Intra-Industry Trade** | Intra-industry trade is defined as the trade of similar products belonging to the same industry or product category. The term is commonly applied to international trade theories, where the same types of goods or services are both imported and exported. |
| **Logistics Costs** | Logistics cost refers to freight charges, warehousing costs, costs of loading and unloading, insurance premiums, and interest charges while goods are in transit. |
| **Environmental Standards** | An environmental standard refers to levels of air pollution, water pollution, thermal pollution and pollution resulting from garbage disposal that a nation allows in International Trade. |

## 4.10 SOME USEFUL REFERENCES


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## 4.11 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

### Check Your Progress 1

1) Read Section 4.2
2) Read Section 4.2
3) Read Section 4.3

### Check your progress 2

1) Read Section 4.4
2) Read Section 4.5

### Check your progress 3

1) Read Section 4.6
2) Read Section 4.7
3) Read Section 4.7